Patrick Thangarajah

MECHANICAL DESIGN | TESTING & VALIDATION | PROTOTYPING | SIMULATION

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Summary

Recent MASc graduate in mechanical engineering with hands-on experience designing, building, and verifying hardware against requirements from concept through validation. Research background in occupant restraint crashworthiness, physical test correlation, DOE, and technical reporting.

Skills

Design & Prototyping	SolidWorks, Onshape, Engineering Drawings, 3D Printing, Lathe, Mill, Bandsaw, Drill Press
Testing & Analysis	DOE, JMP, Minitab, MATLAB, Microsoft Excel, Troubleshooting
Controls & Instrumentation	Arduino, Sensors, Motors, Data Acquisition, Circuit Design, Altium, LTspice
Simulation & Programming	FEA, LS-DYNA, LS-PrePost, Python, C/C++

Education

Master of Applied Science in Mechanical and Mechatronics Engineering — *Waterloo, ON, Canada*
University of Waterloo — *Sep 2023 - Jul 2026*
- Research in occupant restraint crashworthiness and injury biomechanics

Bachelor of Applied Science in Mechanical Engineering, Co-op Program — *Vancouver, BC, Canada*
University of British Columbia — *Sep 2018 - May 2023*
- Biomechanics and Medical Devices Option

Experience

Graduate Research Student - Impact Mechanics and Material Characterization Group — *Waterloo, ON, Canada*
University of Waterloo — *Sep 2023 - Jul 2026*
- Ran a 104-configuration LS-DYNA DOE and used JMP to quantify geometry effects on occupant response, identifying a configuration with 16% lower peak chest deflection
- Validated simulations against physical tests across two configurations and 10 response channels, achieving a 0.71 average ISO 18571 correlation score
- Troubleshot model-to-test discrepancies in positioning, restraint fit, and response timing to improve agreement with physical tests
- Automated seat belt fitting with LS-PrePost scripts, reducing repeated setup steps by 43% (7 to 4)
- Communicated findings through a master's thesis, conference paper, technical reports, and six sponsor presentations

R&D Engineer (Part-time) — *Remote*
Echoscope — *May 2023 - Sep 2023*
- Designed and iterated the bladder testing platform PCB, integrating pressure sensing, Arduino connections, power distribution, and motor and valve interfaces; refined component placement and routing and completed design-rule checks
- Authored five manufacturing and verification procedures covering PCB assembly, wiring, software setup, and functional testing; all five were approved and used by the client

Mechanical Engineering Intern — *Airdrie, AB, Canada*
GN Corporation — *Jan 2022 - Apr 2022*
- Developed standardized work instructions and SOPs by reviewing engineering drawings, customer specifications, and existing manufacturing processes
- Reduced fiberglass composite cutting waste by 50% through process analysis and proposed a material-handling workflow projected to reduce pick-and-place time by 50%

Engineering Intern — *Victoria, BC, Canada*
Breathe Global — *Jul 2021 - Aug 2021*
- Redesigned the analog control circuit for a negative-pressure breathing device in Altium, using LTspice simulation to select low-cost components
- Expanded the device's achievable breathing range from 70% to 88% of target

Projects

Effect of Seat Foam Properties on Injury Response in Rear Impact (ME 725 Course Project) — *Waterloo, ON, Canada*
University of Waterloo — *Jan 2024 - Apr 2024*
- Verified five polyurethane foam material models in LS-DYNA against input stress-strain response using single-element simulations, then implemented them in a rear-impact sled model with a Hybrid III ATD to calculate HIC15 and Nij injury metrics

Soft Distensible Bladder Model - Capstone Project — *Vancouver, BC, Canada*
University of British Columbia — *Sep 2022 - Apr 2023*
- Prototyped bladder phantoms using silicone and agar, comparing pressure-volume response with literature to guide material selection
- Designed and built the control and data-acquisition interface for a pressurized soft-bladder test system, enabling real-time pressure/volume monitoring and automated filling/voiding cycles
- Verified the system against defined performance requirements, demonstrating stable, repeatable operation up to 4.75 kPa internal pressure

Conference Presentations and Publications

- **Thangarajah, P.** (2026). *Assessment of seat belt anchor position on rear seat occupant kinematics in frontal impact using a validated THOR-5F finite element model*. Master of Applied Science thesis, University of Waterloo, Waterloo, ON, Canada. hdl.handle.net/10012/23684
- **Thangarajah, P.**, Watson, B., Tang, K., Tylko, S., & Cronin, D. (2025). *Analysis of rear seat shoulder belt anchor position on kinematics of small-stature female occupants in frontal impact*. Injury Biomechanics Symposium, The Ohio State University, Columbus, OH, USA, May 18–21, 2025.